

Martin Wu

martinwu500@gmail.com | martinwu.tech | [LinkedIn](#) | [GitHub](#)

Education

University of British Columbia

Bachelor of Applied Science in Computer Engineering

Expected May 2029

Vancouver, BC

- Seeking co-op: Summer 2027, 4 or 8 months

Technical Skills

Languages: C, C++, Go, Python, TypeScript, JavaScript

Embedded: STM32, FreeRTOS, SPI, CAN bus, HAL, CMake

Software: gRPC, React, Git, Docker, Vite, Playwright

Experience

Co-Founder & Software Engineer

Oct 2025 – Present

Supercluster

Vancouver, BC

- Co-founded a startup building a Go runtime that lets AI agents reach company databases and APIs only through a policy-checked gateway: an mTLS gRPC control plane dispatches to an isolated data plane, and every request lands in a hash-chained audit log (metadata only, never row data)
- Built `data-plane-local`, a `go run` replica of the production 11-step request dispatcher (identity gate, idempotent writes, audit before any backend call, fail-closed denials) so the team can test the full path without Docker
- Shipped the operator console (HTML/CSS/JS): a live ledger of allowed and denied agent requests, DAG workflows that halt on a denial, and 11 data-source adapters

Embedded Systems Engineer

Sep 2025 – Present

UBC Solar

Vancouver, BC

- Brought up firmware for the Motor Drive Interface board on the Cascadia race car (STM32, FreeRTOS, SPI, CAN filters); 6 merged PRs to `firmware_v4`
- Wrote a MAX31865 driver for PT1000 temperature sensors over 3-wire SPI, verified on hardware to within 1 °C; faulted samples are debounced and never sent on CAN
- Parsed driver-display motor commands off CAN and cached them with a timeout for the motor controller
- Shipped steering-wheel firmware (periodic CAN frames for regen, cruise and lights; AS1115-driven LED speed display) and the external-lights firmware for the v4 car

Projects

Songless | *React, TypeScript, Vite, Cloudflare R2, Web Audio* | Live

2026

- Song-intro guessing game with 100k+ visits: Web Audio clues from 0.01 s to 15 s, then the full track streamed from R2
- Built the media pipeline: source audit, silence trimming, encoding and a hard R2 storage cap; 120-song catalogue

Us | *Phaser 4, TypeScript, Vite, Playwright* | Live

2026

- Browser story game on a custom Phaser engine: grid movement, save/fast travel, dialogue with rewind, 15+ scenes
- Playwright visual-audit tests check lighting, walk cycles and collision so a scene cannot silently regress

Useful Tool Hub | *JavaScript, Python, Vercel* | Live

2025 – 2026

- Ten tools in one page (conversion, PDF, QR, image editing); converters run on-device, downloaders on Vercel Python

Awards

Mathematics (2023–2025): CSMC Top 0.5% nationally (71st) | AIME Qualifier 2x | Euclid Top 2% (85/100)